

Abir Das

Full Stack Developer | React | Node.js | PostgreSQL | Django
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SUMMARY

Backend-focused Software Engineer with experience designing scalable REST APIs, authentication systems, RBAC, PostgreSQL databases, Dockerized deployments, and real-time applications. Experienced across Node.js and ASP.NET Core with a growing focus on AI-powered applications and distributed backend systems.

SKILLS

Languages: JavaScript, TypeScript, C#, Python, SQL

Frontend: React, Angular, Next.js, TypeScript, Tailwind CSS

Backend: Node.js, Express.js, ASP.NET Core, REST APIs, JWT, Socket.IO

Database: MongoDB, PostgreSQL, SQL, Redis, pgvector

Data / ML: Python, NumPy, Pandas, Matplotlib

DevOps: Git, GitHub, Docker, Kubernetes

Concepts: Authentication, RBAC, System Design, Microservices

PROJECTS

Computer Vision-Based Hiragana Character Identifier - ML / CNN / OpenCV June, 2026 - Present

- Implementing an image preprocessing, data augmentation, and model training pipelines to accurately identify Japanese characters from image inputs.

MathCity - AI Ready Learning Platform June, 2026 - Present

- Built a scalable learning management platform supporting Subjects, Topics, Lessons, Resources and Practice Questions.

Student-College Guidance App - React, Node.js, Redux, MongoDB Clusters, Agentic AI. Oct, 2025

- A platform that connects an **user with Interests and percentages** to ideal streams with **AI agent**.
- Suggested **registered colleges** based on student profile and admission requirements.
- Integrated **MongoDB** for data storage and **Context API** for global state management.
- Used **API.postalpincode** for location-based filtering and recommendations.

Real-Time Chat App - React, Socket.IO, Express.js, MongoDB, JWT. Sep, 2025

- A Chat application with **OTP login** and **JWT authentication**.
- Implemented **real-time messaging**, **notifications**, and **group chat** using **Socket.IO**.
- Added **follow/unfollow functionality**, **user filtering by location**, and **autocomplete location search**.
- Integrated **Dicebear avatars** for dynamic user profile visuals.

NASA Space Data Explorer - React, Node.js, JWT, Context API, Nodemailer. - <https://ninjanights.github.io/nasahairclips/> Aug, 2025

- Developed a full-stack NASA Explorer using React, Node.js, Express, and JWT with secure APIs, APOD integration, space weather tracking, and automated email services.

Pokémon Explorer - React, Custom Hooks, Prop drilling. - pokemonforest.vercel.app Jul, 2025

- Developed a React-based Pokémon Explorer integrating PokeAPI to display Pokémon stats, abilities, evolutions, and interactive location mapping using custom hooks.

WORK EXPERIENCE

Dec, 2025 - June, 2026

- Designed and implemented Full-stack applications including a Multi-Tenant POS System, Payroll and Hotel Management Web & Native App, and Job/Product Posting Platform using React, Node.js, PostgreSQL, Redis, pgvector, Docker.
- A system for selling softwares, booking appointments, job posting/hirings, vector searched chat/q&a.
- Architected and developed a **multi-tenant Smart Agriculture ecosystem** serving administrators and farmers across mobile and web platforms.
- Designed scalable backend services using **Node.js, PostgreSQL, Redis, pgvector, Docker, and REST APIs** to support inventory, crop management, AI search, and real-time operations.
- Implemented semantic search using **pgvector embeddings**, enabling farmers to chat with an AI assistant about crop diseases, fertilizer recommendations, irrigation schedules, and crop growth stages.
- Built an integrated agricultural dashboard combining **weather forecasts, soil insights, crop health monitoring, irrigation recommendations, and live Indian mandi prices** into a single decision-support system.
- Developed an e-commerce module supporting inventory management, product catalog, order processing, and agricultural product distribution.
- Implemented secure authentication, authorization, and real-time GPS tracking for registered farmers across mobile and web applications.

Education: Higher Secondary | **Language:** English (Fluent) | Illustrator / Designer